Lecture 12: Mesh Current Analysis

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current, mesh current 1 voltage supermesh

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T \mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n \mathbf{e} = \mathbf{i}_n$.

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- From the ground to node k: e_k is known.
- Between two nodes: a supernode.
- A dependent source: one more constraint.

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Can we do the opposite?

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
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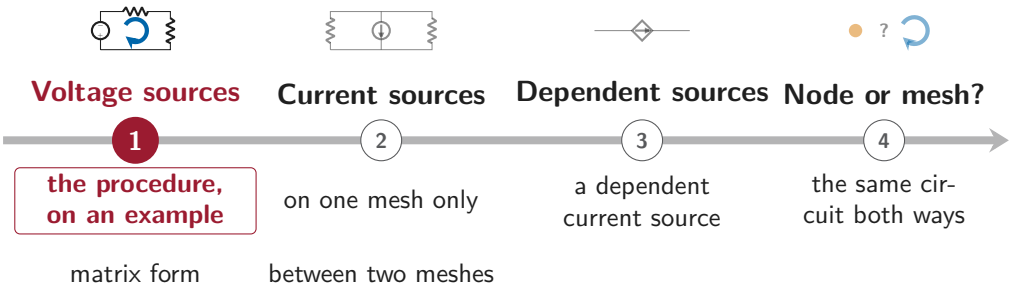
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- Meshes need a planar circuit.

Mesh current analysis follows the same plan, with KCL and KVL swapped.

Roadmap

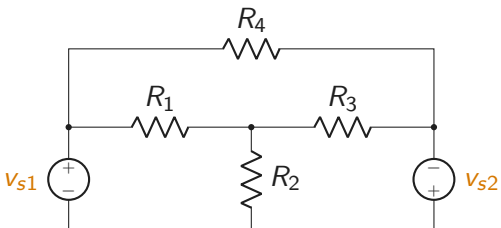


Each station adds a new kind of source; the last one compares the two methods.

Example 12.1: mesh current analysis step by step

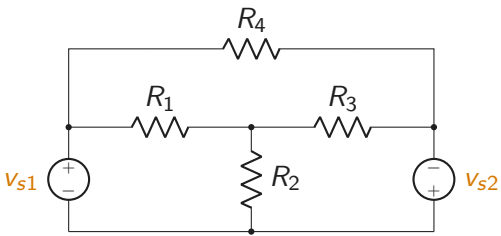
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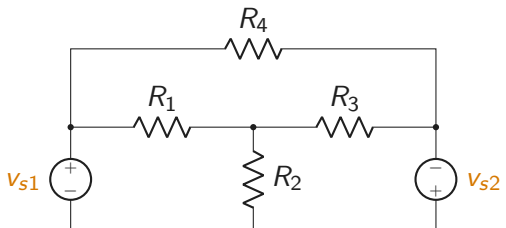


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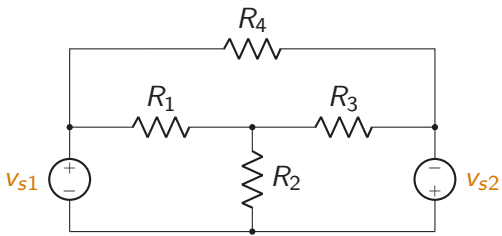
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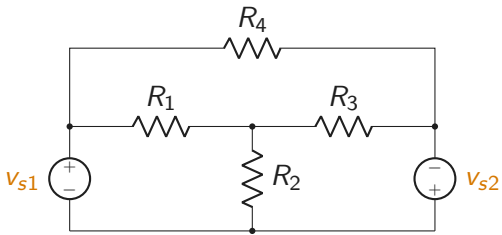
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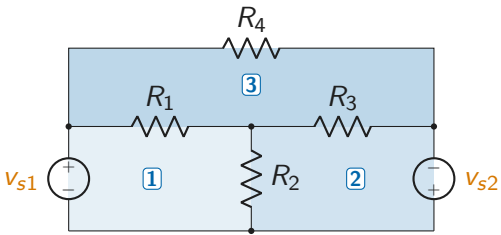
We find them in four steps, on this circuit.

Step 1: find the inner meshes



The circuit is planar: it is drawn with no crossing branches.

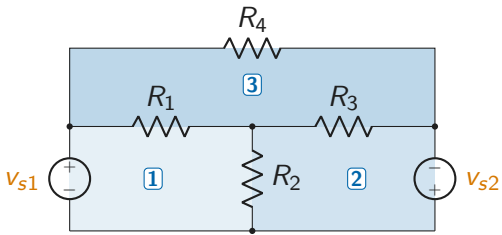
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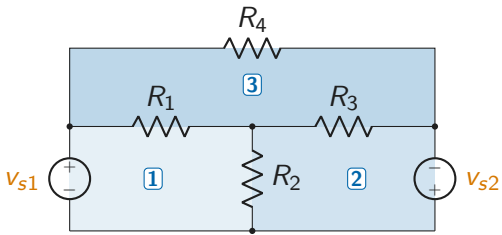


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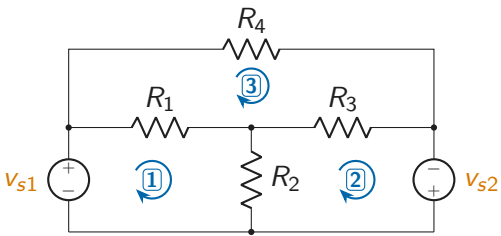
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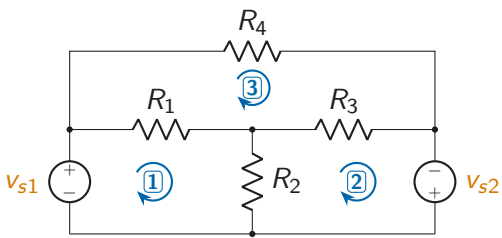
Mesh current analysis has one unknown per inner mesh.

Step 2: assign the mesh currents



Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

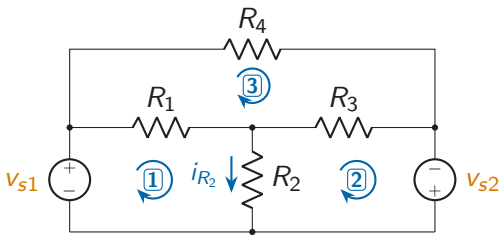
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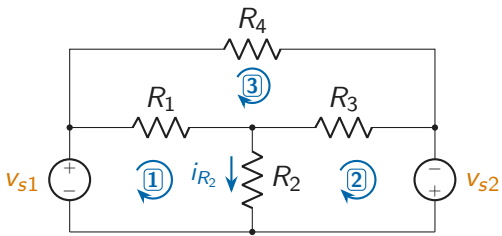
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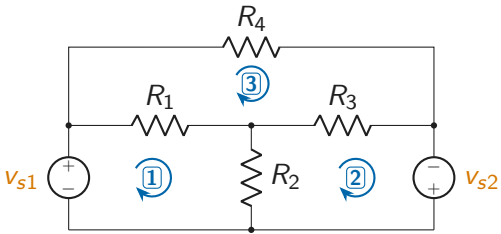
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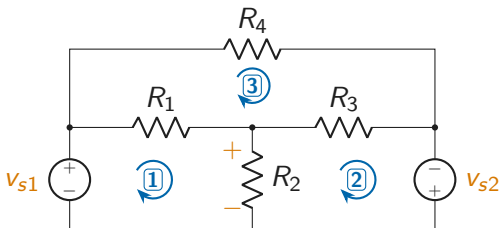
A mesh current enters and leaves every node it passes, so KCL holds automatically.

Step 3: write the resistor voltages with mesh currents



Ohm's law gives each resistor voltage from its mesh currents.

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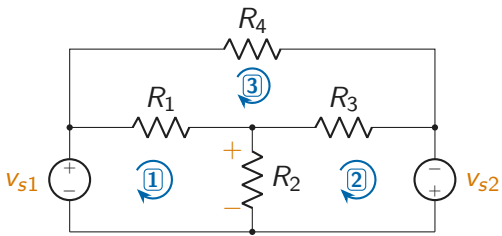


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The voltage of R_2 along mesh 1 (top to bottom):

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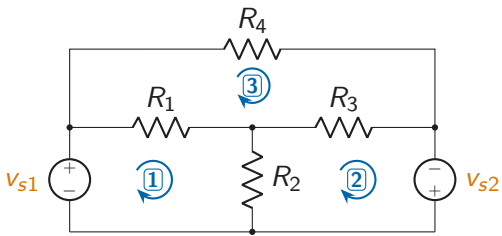
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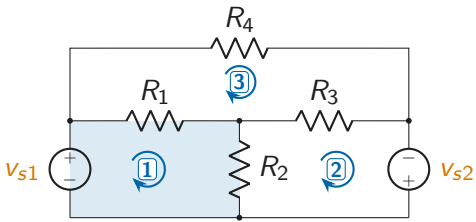
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The rule: R times (own mesh current minus the other one).

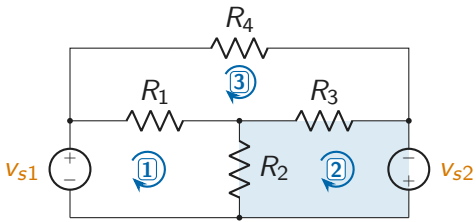
Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

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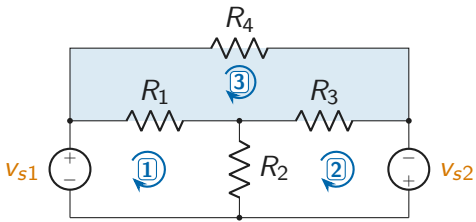
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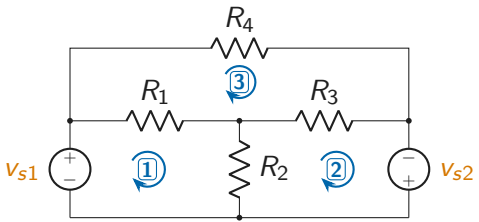
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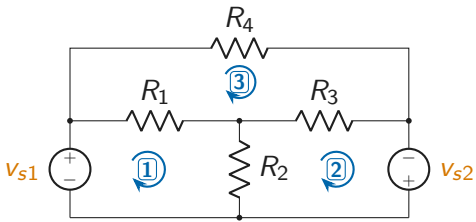
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A source met from $-$ to $+$ enters with a minus sign.

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Three equations, three unknowns.

Then Steps 2 and 3 give every branch current and voltage.

Roadmap



Each station adds a new kind of source; the last one compares the two methods.

Example 12.1 in matrix form

Collect the terms of each mesh:

$$\begin{aligned}(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\ -R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\ -R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0\end{aligned}$$

Question. Write the three equations as one matrix equation.

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Z_M is the mesh impedance matrix: $Z_M \mathbf{i}_M = \mathbf{v}_M$.

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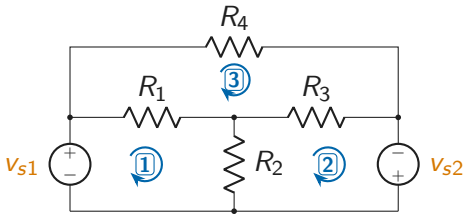
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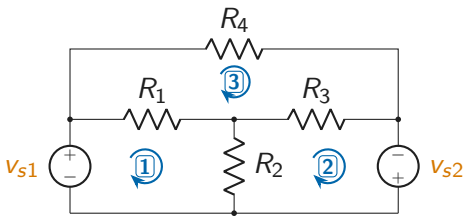
$\mathbf{v}_M$: the source voltages rising along each mesh.

Writing Z_M directly from the circuit: three rules



Question. Can we read Z_M from the circuit, without writing KVL?

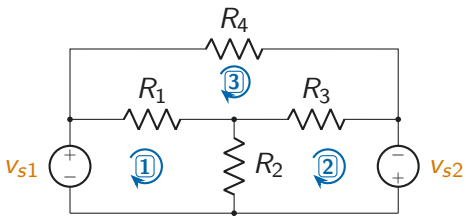
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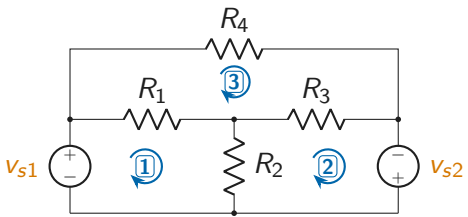


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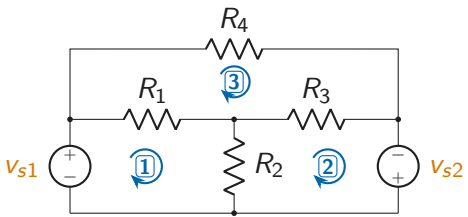
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Right side k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

Writing Z_M directly from the circuit: three rules



Question. Can we read Z_M from the circuit, without writing KVL?

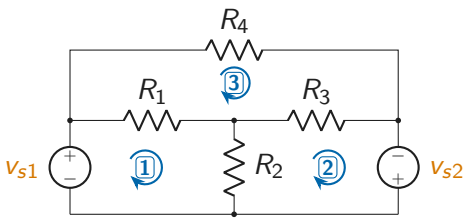
Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

Right side k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

So Z_M is symmetric: entry (k, j) equals entry (j, k).

Writing Z_M directly: Example 12.1

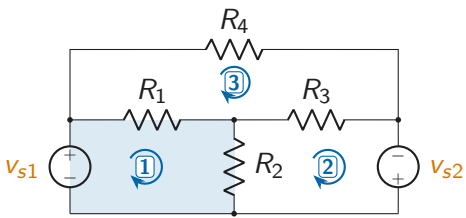


$$Z_M = \begin{bmatrix} & \\ & \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} \\ \end{bmatrix}$$

Question. Fill in Z_M and $\mathbf{v}_M$ from the circuit.

Writing Z_M directly: Example 12.1

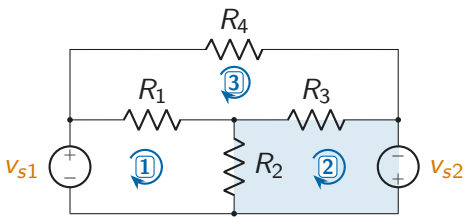


$$Z_M = \begin{bmatrix} R_1 + R_2 & & \\ & & \\ & & \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

Diagonal $(1,1)$: around mesh 1 are R_1 and R_2.

Writing Z_M directly: Example 12.1

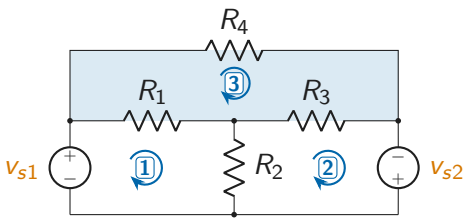


$$Z_M = \begin{bmatrix} R_1 + R_2 & & \\ & R_2 + R_3 & \\ & & \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

Diagonal $(2,2)$: around mesh 2 are R_2 and R_3.

Writing Z_M directly: Example 12.1

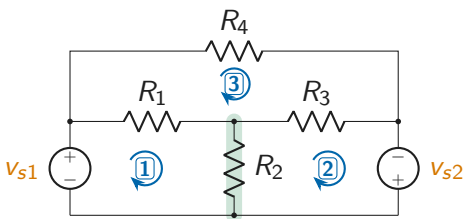


$$Z_M = \begin{bmatrix} R_1 + R_2 & & \\ & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} \\ \\ \end{bmatrix}$$

Diagonal $(3,3)$: around mesh 3 are R_1, R_3 and R_4.

Writing Z_M directly: Example 12.1

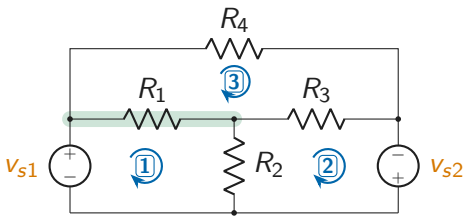


$$Z_M = \begin{bmatrix} R_1 + R_2 & -R_2 & \\ -R_2 & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} \\ \\ \end{bmatrix}$$

Meshes 1 and 2 share R_2: entries $(1,2)$ and $(2,1)$ are $-R_2$.

Writing Z_M directly: Example 12.1

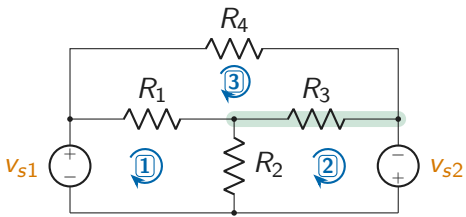


$$Z_M = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & \\ -R_1 & & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} \\ \\ \end{bmatrix}$$

Meshes 1 and 3 share R_1: entries $(1,3)$ and $(3,1)$ are $-R_1$.

Writing Z_M directly: Example 12.1

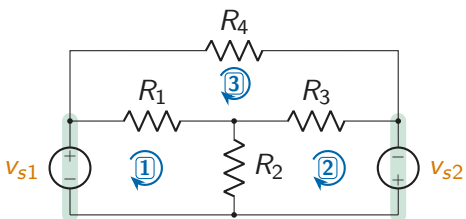


$$Z_M = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Meshes 2 and 3 share R_3: entries $(2,3)$ and $(3,2)$ are $-R_3$.

Writing Z_M directly: Example 12.1

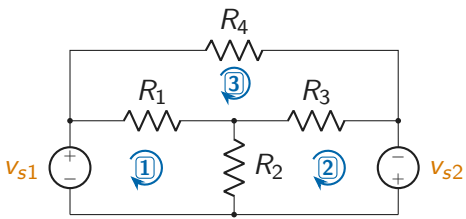


$$Z_M = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Right side: mesh 1 meets v_{s1} from $-$ to $+$, mesh 2 meets v_{s2} from $-$ to $+$, mesh 3 has no source.

Writing Z_M directly: Example 12.1

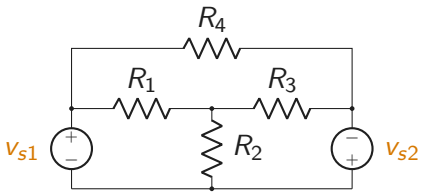


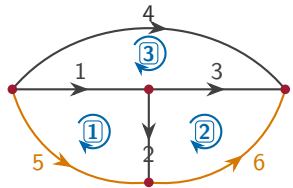
$$Z_M = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

$$\mathbf{v}_M = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

The same Z_M as from KVL ✓, and it is symmetric.

Z_M from the graph: the mesh matrix

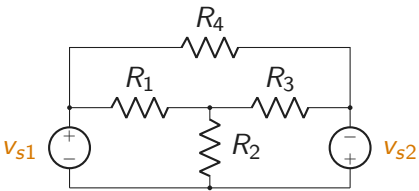


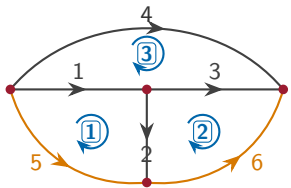


Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

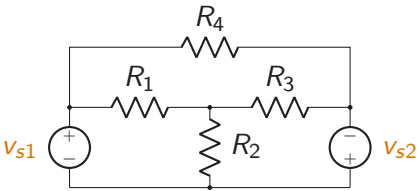
Question. Write the mesh matrix M.

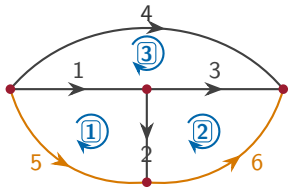
One row for each inner mesh.

Entry $+1$: the branch goes along the mesh; -1: against it.

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

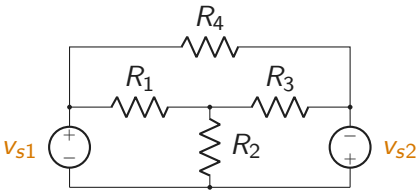
One row for each inner mesh.

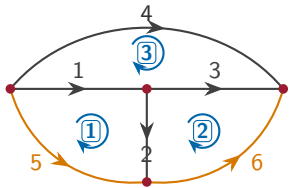
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

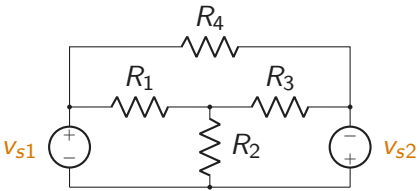
One row for each inner mesh.

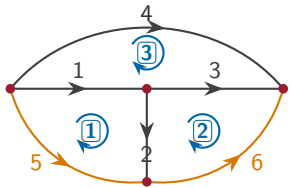
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

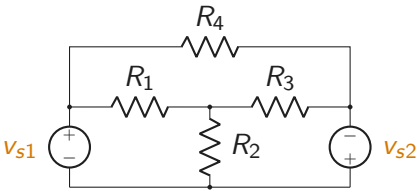
One row for each inner mesh.

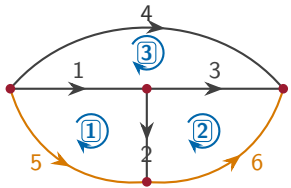
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Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

One row for each inner mesh.

Entry $+1$: the branch goes along the mesh; -1: against it.

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccc|cc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{array}\right] \end{array}$$

M_R: resistors M_S: sources

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M : \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

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b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M: \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Resistors: $\mathbf{v}_R = R\,\mathbf{i}_R$, $R = \mathrm{diag}(R_1, \ldots, R_{b_R})$, one resistance per resistor.

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

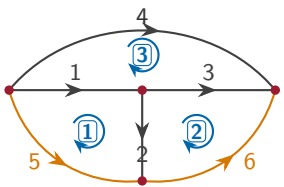
$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

$Z_M^T = M_R\, R^T M_R^T = M_R\, R\, M_R^T = Z_M$, since R is diagonal.

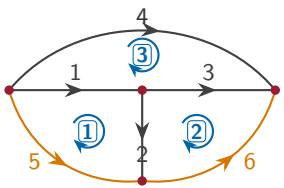
So Z_M is always symmetric for resistors and voltage sources.

Z_M from the graph: Example 12.1, the resistor currents

M_R^T: the resistor columns 1–4 of M, written as rows.



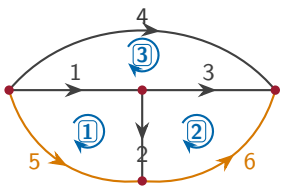
Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

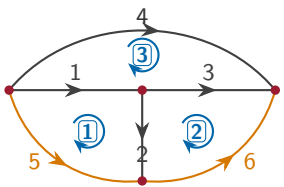
Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Z_M from the graph: Example 12.1, the resistor currents

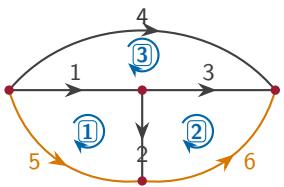


M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Each resistor current: the mesh currents along its arrow minus those against it.

Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Each resistor current: the mesh currents along its arrow minus those against it.

Then Ohm's law: $\mathbf{v}_R = R\,\mathbf{i}_R$, $v_{R_k} = R_k i_{R_k}$.

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

Z_M from the graph: Example 12.1, the matrix

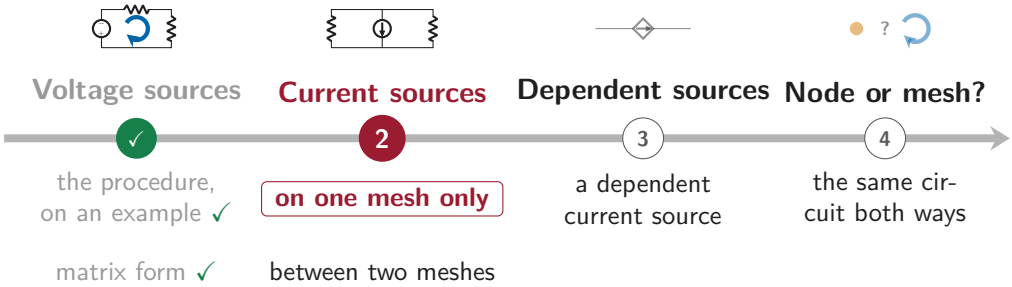
$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

The three rules read this product off, entry by entry.

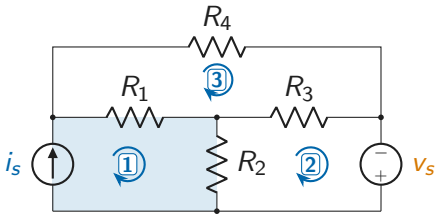
Roadmap



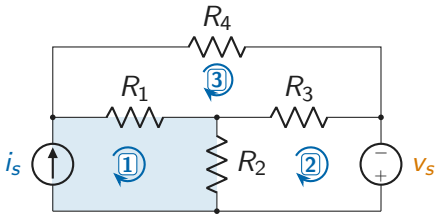
Each station adds a new kind of source; the last one compares the two methods.

Example 12.2: a current source on one mesh only

i_s is on mesh 1 only: $i_{M1} = i_s$ is known.



Example 12.2: a current source on one mesh only

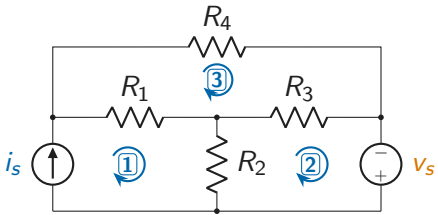


i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Example 12.2: a current source on one mesh only



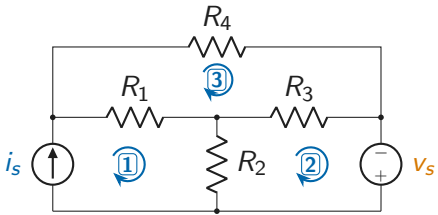
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Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Example 12.2: a current source on one mesh only



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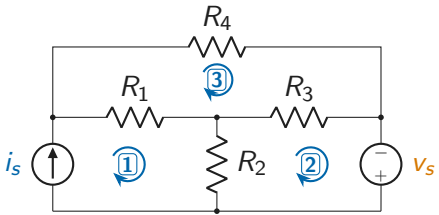
The unknowns are i_{M2}, i_{M3}.

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

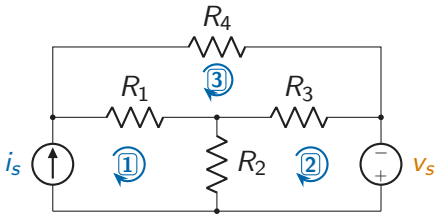
Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

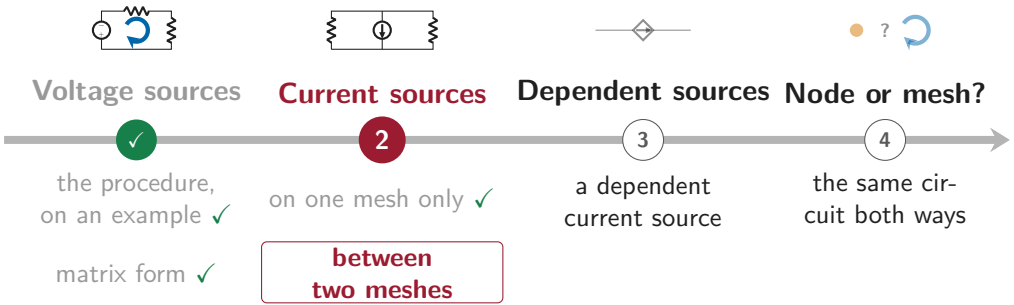
Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

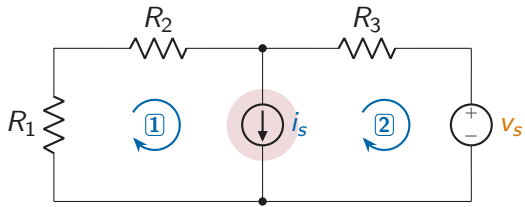
The known i_s moves to the right side.

Roadmap



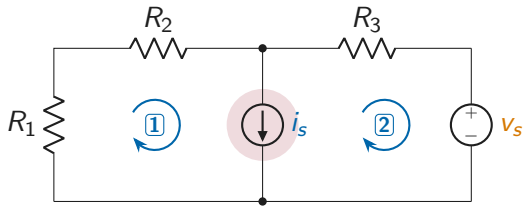
Each station adds a new kind of source; the last one compares the two methods.

Example 12.3: a current source between two meshes



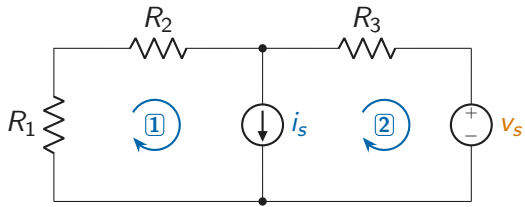
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?

Example 12.3: a current source between two meshes



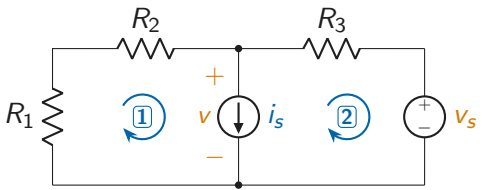
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.

Example 12.3: a current source between two meshes



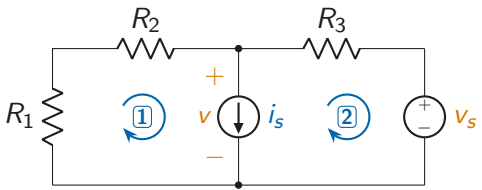
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.
- **Modified mesh current analysis** offers two ways out: Method I (its voltage is a new unknown) and Method II (a supermesh).

Method I: the voltage as an unknown



The voltage v of i_s is one more unknown.

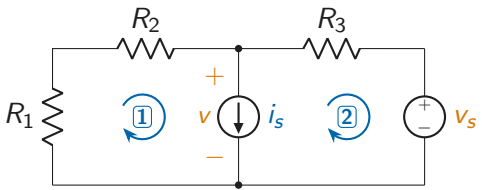
Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown

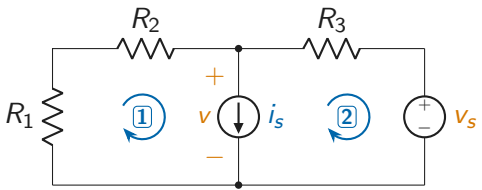


Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown



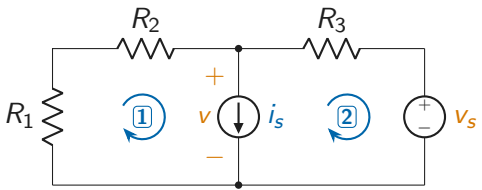
Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

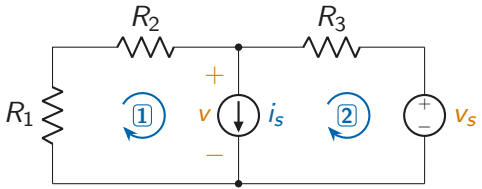
Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Method I: the voltage as an unknown



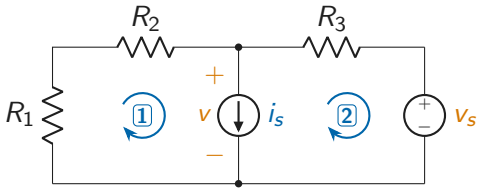
$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Three unknowns, three equations; the last row is the constraint.

Method I: the voltage as an unknown



$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

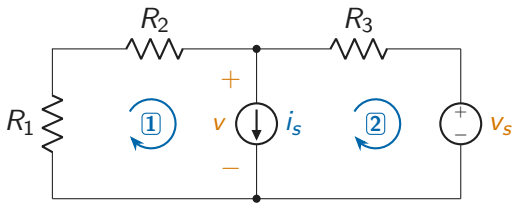
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Three unknowns, three equations; the last row is the constraint.

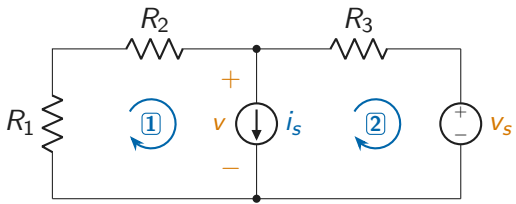
It works, but the supermesh avoids the extra unknown.

Method II: the supermesh



Question. Can we get rid of the unknown v?

Method II: the supermesh

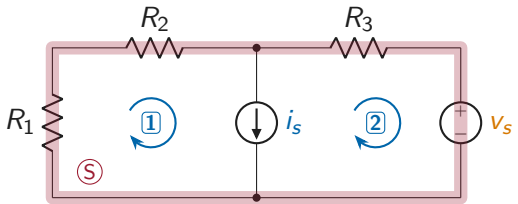


Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

Method II: the supermesh



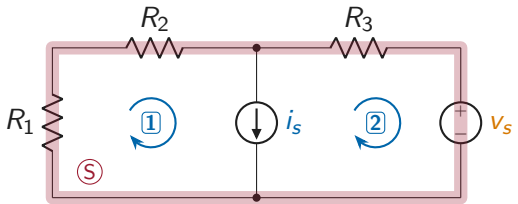
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v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Method II: the supermesh



Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

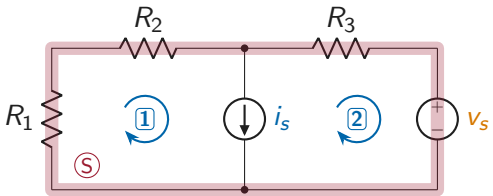
v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Supermesh

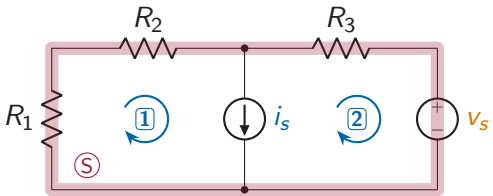
Write KVL around two meshes as if they were a single mesh, skipping the current source between them.

Method II: the equations



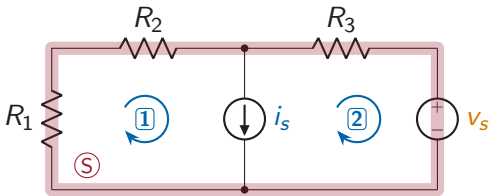
Question. Write KVL around the supermesh.

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

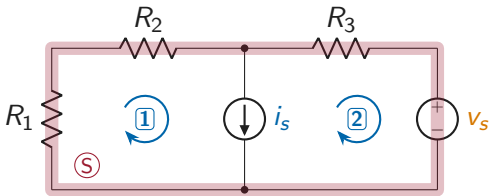
Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method II: the equations

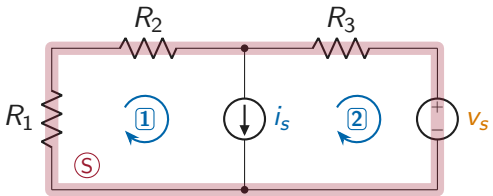


Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Method II: the equations



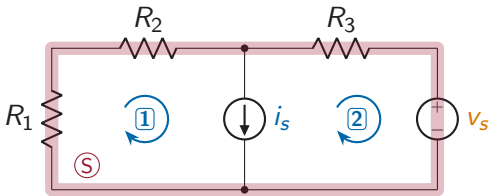
Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

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Two unknowns: one fewer than Method I.

Method II: the equations



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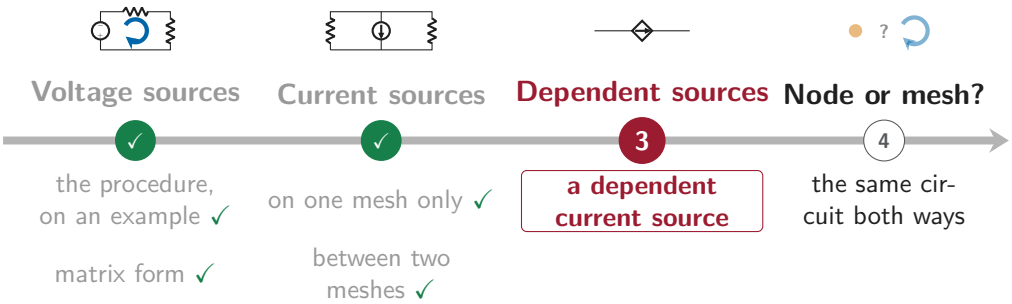
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The matrix is not symmetric any more.

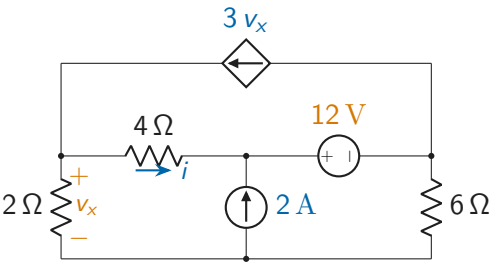
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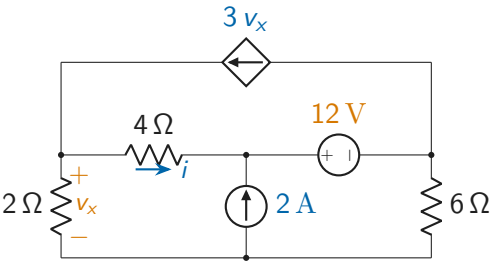
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Example 12.4: a dependent current source

Question. Find i, the current of the $4\,\Omega$.



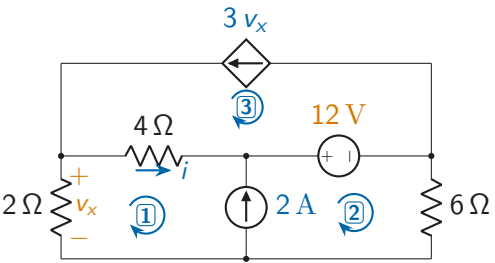
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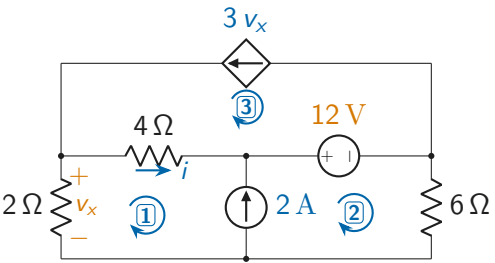


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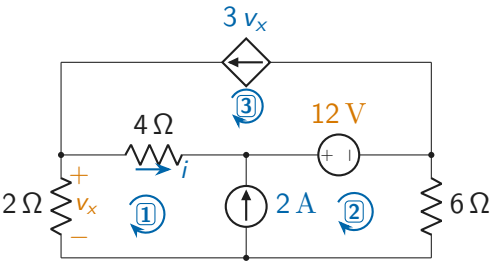
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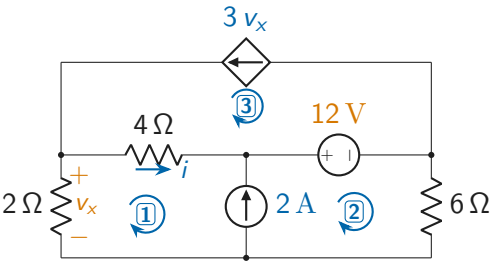
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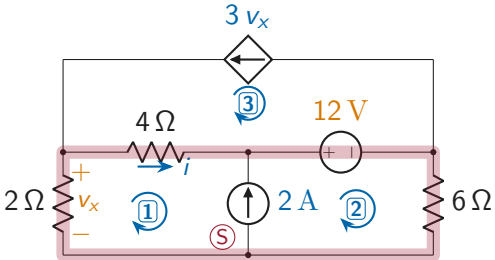
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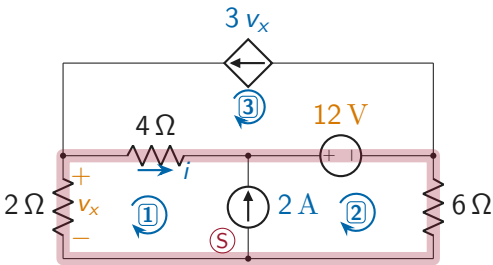
Equations: KVL around the supermesh and two constraints.

Example 12.4: the equations



Question. Write KVL around the supermesh.

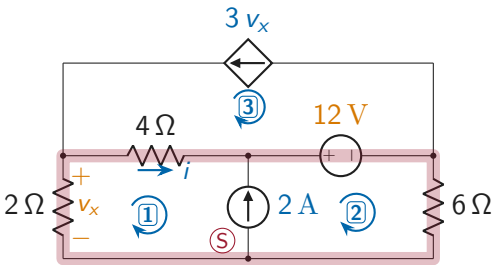
Example 12.4: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

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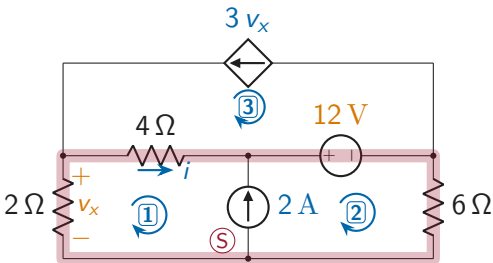


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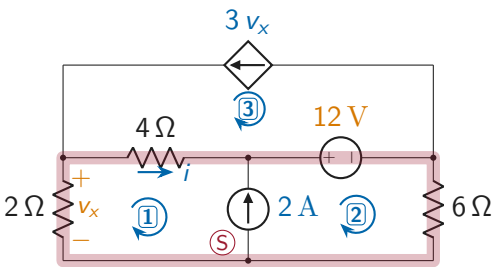
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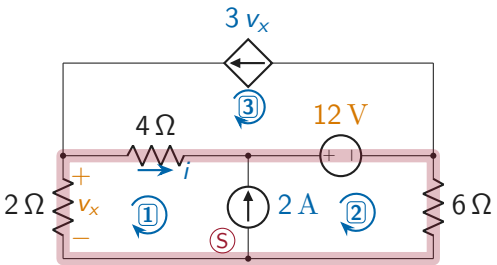
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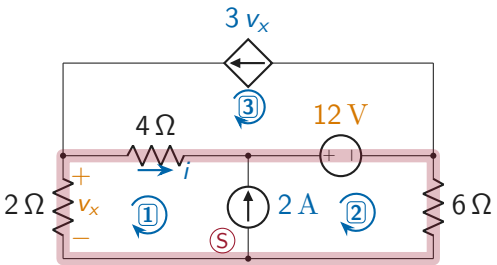
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Constraint: $i_{M3} = 6\,i_{M1} \quad (3)$

Example 12.4: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

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Solve: $i_{M1} = 2\,\mathrm{A}$, $i_{M2} = 4\,\mathrm{A}$, $i_{M3} = 12\,\mathrm{A}$.

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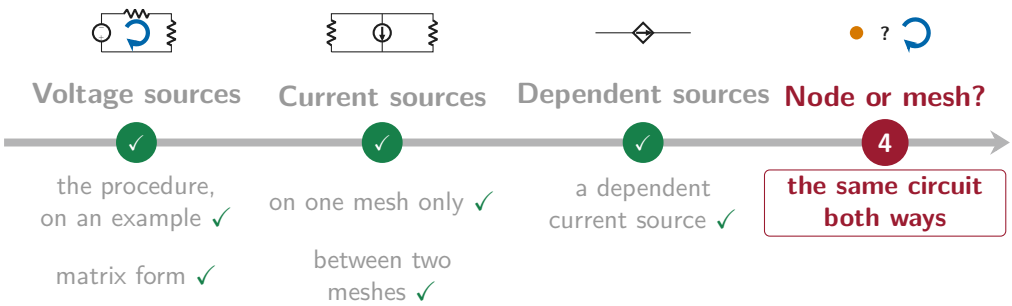
- A dependent source is handled like an independent one of the same kind.
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Remark

If a dependent current source is on mesh k only, do not write KVL around mesh k: its voltage is not known.

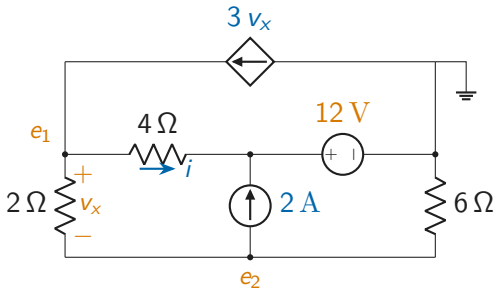
Write the constraint that the dependent source gives instead.

Roadmap



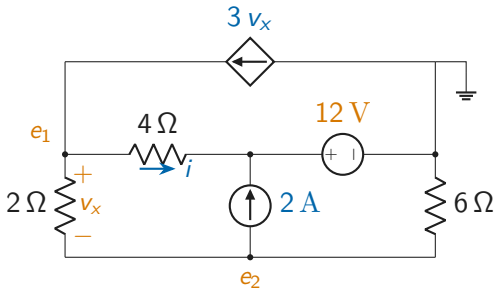
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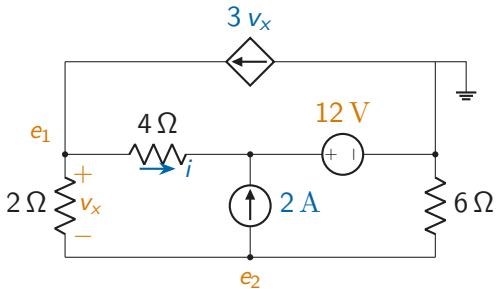
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Take the right node as the ground.

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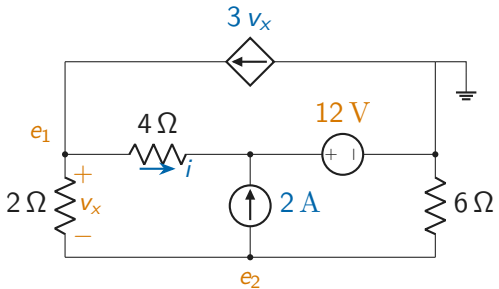
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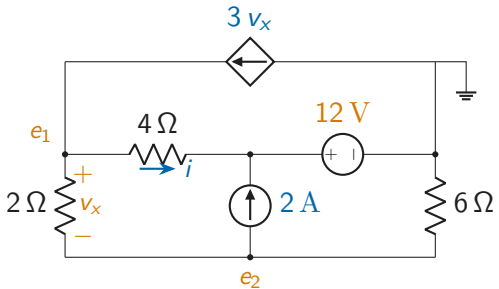
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Constraint: $v_x = e_1 - e_2$

Example 12.4 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

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The same answer, as it must be.

Node or mesh?

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$n - 1 = 3$ unknowns.

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Tip. Count before you start: mesh, ℓ minus the current sources; node, $n - 1$ minus the voltage sources.

Mesh current analysis needs a planar circuit; node voltage analysis always works.

Summary

In the circuit	What to do
resistors and voltage sources	KVL around every inner mesh: $Z_M \mathbf{i}_M = \mathbf{v}_M$, from three simple rules; $Z_M = M_R R M_R^T$ is symmetric
a current source on one mesh only	that mesh current is known; no KVL around that mesh
a current source between two meshes	Method I: its voltage as an unknown, plus the constraint; Method II: KVL around the supermesh, plus the constraint
a dependent source	write its control variable with mesh currents: a constraint equation
node or mesh?	count the equations; mesh needs a planar circuit

After the mesh currents are found: $\mathbf{i} = M^T \mathbf{i}_M$, then the branch voltages from the element laws.